

Samira Malek

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Summary

Ph.D. candidate in **Computer Science** specializing in **Machine Learning**, with 6+ years of experience in **deep learning, optimization, algorithms, and statistics**. Skilled in developing and deploying **large scale models** for high-impact **AI** applications and technologies.

Education

Pennsylvania State University

Ph.D. in Computer Science

Thesis: Interpretable, Convergent, Deep Neural Networks for Image Restoration.

PA, USA

Expected August 2027

Pennsylvania State University

M.S. in Computer Science

PA, USA

Aug 2025

Sharif University of Technology

M.S. in Electrical Engineering

Tehran, Iran

Sep 2020

Sharif University of Technology

B.S. in Electrical Engineering

Tehran, Iran

Feb 2018

Experience

Pennsylvania State University

Research Assistant

PA, USA

2023 – Present

- Designed and implement an interpretable, convergent **convolutional neural network (CNN)** for **image deblurring** achieving 33.84 dB PSNR/0.94 SSIM, and providing **theoretical convergence proofs**.
- Developed an **mathematically grounded deep neural network (UPADNet)** leveraging Fourier phase information for **image deblurring**, achieving state-of-the-art performance (34.63 dB PSNR, 0.975 SSIM) with improved robustness under noise and low-data regimes.
- Evaluated advanced **large language models** (DeepSeek, Gemini 2.0, O-series) on Olympiad mathematics.
- Created a **multimodal** dataset (1,135 questions) for discrete mathematics reasoning; benchmarked GPT-5, Gemini, Gemma.
- Developed an **agentic AI** system for automated grading of mathematical proofs.
- Skills:** (non)convex optimization, torchvision, Agentic AI, designing and training DNN, Image Processing, Signal Processing, training on distributed computing, data collection, computer vision, NLP

Applied Research Laboratory

Machine Learning Researcher

PA, USA

2024 – 2025

- Built a **hierarchical topic modeling using BERTopic + GPT**, and generating sentence-level descriptions for topics.
- Developed an **LLM-based web application** that detects health misinformation and **generates warning message** with 87% accuracy.
- Skills:** Finetuning generative language models, topic modeling, Prompt engineering, unsupervised clustering, Hugging face, OpenAI API, Agentic AI, RAG

SNOWA

Data & Analytics Engineer

Tehran, Iran

2022 – 2023

- Developed **customer segmentation using unsupervised learning** to **support targeted marketing**.
- Applied graph-based algorithms with SQL to identify and **analyze customer relationship networks**.
- Designed and developed a **scalable Data Warehouse** to **support business intelligence and analytics**
- Skills:** SQL, Python, PCA, Microsoft Power BI, Microsoft SQL Server Management Studio, Scikit-learn, large-scale data engineering

Sharif University of Technology

Research Assistant

Tehran, Iran

2018 – 2020

- Designed **deep neural networks** based on a **message-passing algorithm** for decoding linear codes, achieving BER of 10^{-7} at 6 dB SNR.
- Skills:** TensorFlow, NumPy, Pandas, Linux, Statistics, MAP Estimation, information theory, communication, graph-based models, MATLAB

Selected Publications

- "Combating Health Misinformation with LLM: Automated Detection, Thematic Analysis, Inoculation", **JMIR**, 2026.
- "CombiGraph-Vis: A Curated **Multimodal Olympiad Benchmark** for Discrete Mathematical Reasoning", **Math-AI NeurIPS** 2025.
- "Refgrader: Automated Grading Of Mathematical Competition Proofs Using **Agentic Workflows**", **Math-AI NeurIPS** 2025.
- "Brains vs. bytes: Evaluating **LLM** proficiency in olympiad mathematics", **Conference On Language Modeling (COLM)** 2025.
- "Multi variable-layer **neural networks** for decoding linear codes", **IEEE Workshop on Communication and Information Theory** 2020.
- "A Methodology Framework for Analyzing Health Misinformation to Develop Inoculation Interention **Using Large Language Models**: A case study on covid-19", **AEJMC 108th Annual Conference**.
- "**Data-Driven Model** to Improve Mechanical Harvesters for Nut Trees", **ASABE Annual International Meeting** 2024.
- "A **Large Language Model-Based** Analyzing of HPV and COVID-19 Misinformation on Social Media", under review in **DHJ**.
- "**Convergent Deep Blind Image Deblurring**: A Cascaded Unrolled Networks Approach", under review **IEEE Transactions on Neural Networks and Learning Systems**.
- "Leveraging Phase Information to Boost Unrolled Network Learning for **Image Deblurring**", under review at **ECCV** 2026.

Awards and Honors

- Awarded \$4,000 for **Execlent Research** from Information Science Technology department of Pennsylvania State university 2026

- Awarded \$4,000 from The Elmore Family School of Electrical and Computer Engineering of **Purdue University** 2022
- Awarded \$2,000 from College of Engineering Dean's Office Fellowship of Pennsylvania State university 2022
- Ranked 29th among +40,000 (**Top 0.01%**) in Iran National M.S. Entrance Exam 2018
- Ranked 40th among +250,000 (**Top 0.01%**) in Iran National B.S. Entrance Exam 2013
- **Silver Medalist**, 30th Iran **National Mathematical Olympiad** 2012
- Admission to high school in **National Organization for Development of Exceptional Talents** (success rate < **0.3%**) 2009

Selected Courses

- Deep Learning for NLP; Large-Scale Machine Learning; Machine Learning Algorithms and Tools
- Convex Optimization; Non-Convex Optimization; Distributed Optimization
- Pattern Recognition and Machine Learning; Secure and Robust Machine Learning; AI and Biological Computation
- Probability and Statistics; Stochastic Processes; Speech Processing, Information Hiding
- Communication Systems; Digital Signal Processing; Image Processing

Selected Academic Projects

Deep Learning

- Fallacy Science Question Answering with LLMs** Fall 2024
 - Built dataset of fallacy-based science questions; evaluated Generative AI models (Llama, Gemini, and ChatGPT), improving model accuracy on reasoning by 60% through optimized prompts.
- Identifying the Source Model of Generated Text** Fall 2024
 - Generated datasets using five large language models and trained BERT classifiers to distinguish text by source model.
- Multi-hop Question Answering with GPT** Spring 2023
 - Trained GPT models via prompting for multi-hop reasoning on the HotPotQA benchmark.
- Pitch Tracking in Noisy Speech Signal** Spring 2019
 - Implemented feedforward, recurrent neural networks (RNN), and LSTM based models for pitch estimation in noisy speech signal.
- EEG-based Typing Classification** Spring 2016
 - Designed a neural network to classify right- vs. left-handed typing using EEG signals.
- Backdoor Attack and Detection on CIFAR-10** Spring 2024
 - Designed and evaluated additive backdoor attacks and implemented detection methods for image classifiers.

Optimization

- MinMax Optimization Algorithms** Spring 2023
 - Implemented Stochastic Gradient Descent Ascent (SGDA) and Stochastic Compositional Gradient Descent Ascent (SCGDA) for min-max optimization problems.
- Classical Optimization Methods** Fall 2018
 - Implemented BFGS, Steepest Descent, and Newton's method with line-search for nonlinear optimization.

Signal\Image Processing & Communications

- Viterbi Algorithm and Sequence Detection** Spring 2020
 - Implemented the Viterbi algorithm and developed a reduced-complexity sequence detector for communication systems.
- Speech Signal Pitch Contour Extraction** Spring 2019
 - Extracted pitch contours from speech time series signals using SIFT, HPS, and AMDF algorithms.
- Image Recovery via Sparse Methods** Fall 2018
 - Applied IMAT and OMP algorithms for image recovery from incomplete data.
- Signal and Image Reconstruction Techniques** Fall 2018
 - Reconstructed time series signal and image using SDFT and RS methods.

Teaching Experience

- **TA for Discrete Mathematics** in Pennsylvania State University. Spring & Fall 2023, Spring 2024
- **TA for Signal & System** in Sharif University of Technology. Fall 2019
- **TA for Stochastic Random Process** in Sharif University of Technology. Spring 2019
- **TA & MATLAB Teaching for Engineering Mathematics** in Sharif University of Technology. Spring 2019
- **Design Mock Test** of the Iran Nation-wide University Entrance Exam (Konkoor) for B.S degree in Kanoon Institution, which has the most participants in Iran (more than +20,000 students every year). 2017–2018
- **Teaching Mathematics** Farzanegan High school (NODET), Preparing students for Iran National Mathematical Olympiad. 2014–2015

Skills

Programming & Tools: Python (PyTorch,Tensorflow) , MATLAB, SQL, C, R, HTML, Git, Linux, GPU/CUDA